F78AB - Actuarial and Financial Mathematics B

Tutorial 5 – Nominal interest rates

We refer to an annuity which is payable m times per year as *payable m-thly*.

Suppose we wish to value a level annuity which is payable m-thly in arrears for n years, with the total annual payment £1.

For example, if $m = 12$ and $n = 5$, then the annuity pays £$\frac{1}{12}$ at the end of each month for 5 years.

$s^{(m)}_{\overline{n}\rceil}$ represents its value at the end of the payment period and $a^{(m)}_{\overline{n}\rceil}$ represents its value at the start of the payment period.

If we wish to emphasize the interest rate per unit time period that is used in the calculation, we write $s^{(m)i}_{\overline{n}\rceil}$ or $a^{(m)i}_{\overline{n}\rceil}$, as appropriate.

Question 1

1. Prove that
$$a^{(m)}_{\overline{n}\rceil} = \frac{1 - v^n}{i^{(m)}}.$$

2. Hence show that
$$s^{(m)}_{\overline{n}\rceil} = \frac{(1+i)^n - 1}{i^{(m)}}.$$

Question 2

1. State in words the meaning of the notation $\ddot{s}^{(12)}_{\overline{10}\rceil}$.

2. State in words the meaning of the notation $600\ddot{a}^{(12)}_{\overline{10}\rceil}$.

3. Prove that
$$\ddot{s}^{(m)i}_{\overline{n}\rceil} = \frac{(1+i)^n - 1}{d^{(m)}}.$$

4. Hence show that
$$\ddot{a}^{(m)}_{\overline{n}\rceil} = \frac{1 - v^n}{d^{(m)}}.$$

Question 3 Evaluate the following.

1. $s^{(2)\,0.05}_{\overline{10}\rceil}$ 2. $a^{(6)\,0.1}_{\overline{17}\rceil}$ 3. $\ddot{s}^{(4)\,0.035}_{\overline{25}\rceil}$ 4. $\ddot{a}^{(12)\,0.12}_{\overline{7}\rceil}$ 5. $\ddot{a}^{(4)\,0.06}_{\overline{\infty}\rceil}$

Numerical solutions: Q3.1. 12.73320; Q3.2. 8.34959; Q3.3. 39.79822; Q3.4. 4.85527; Q3.5. 17.28711.

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Solutions to Tutorial 5 – Nominal interest rates

Solution 1

1. Define $j := (1+i)^{1/m} - 1$. Then

$$a_{\overline{n}|}^{(m)i} = \frac{1}{m}\, a_{\overline{nm}|}^{j} = \frac{1}{m}\,\frac{1-(1+j)^{-nm}}{j} = \frac{1}{m}\,\frac{1-(1+i)^{-n}}{(1+i)^{1/m}-1}.$$

   As

$$1+i = \left(1+\frac{i^{(m)}}{m}\right)^m \quad \Rightarrow \quad i^{(m)} = m\cdot\left[(1+i)^{1/m}-1\right],$$

   then by substitution for $i^{(m)}$,

$$a_{\overline{n}|}^{(m)i} = \frac{1-(1+i)^{-n}}{i^{(m)}} = \frac{1-v^n}{i^{(m)}}.$$

2. As

$$s_{\overline{n}|}^{(m)} = (1+i)^n\, a_{\overline{n}|}^{(m)},$$

   it follows immediately from the first part that

$$s_{\overline{n}|}^{(m)} = \frac{(1+i)^n - 1}{i^{(m)}}.$$

Solution 2

1. $\ddot{s}_{\overline{10}|}^{(12)}$ denotes the accumulation at the end of 10 years of a level annuity of £$\frac{1}{12}$ paid at the start of each month for 10 years.

2. $600\ddot{a}_{\overline{10}|}^{(12)}$ denotes the present value of a level annuity of £50 paid at the start of each month for 10 years.

3. Prove that

$$\ddot{s}_{\overline{n}|}^{(m)i} = \frac{(1+i)^n - 1}{d^{(m)}}.$$

   Define $j := (1+i)^{1/m} - 1$. Then

$$\ddot{s}_{\overline{n}|}^{(m)i} = \frac{1}{m}\,\ddot{s}_{\overline{nm}|}^{j} = \frac{1}{m}\,\frac{(1+j)^{nm}-1}{\frac{j}{1+j}} = \frac{1}{m}\,\frac{(1+i)^n-1}{1-(1+i)^{-1/m}}.$$

   As

$$1-d = \left(1-\frac{d^{(m)}}{m}\right)^m \quad \Rightarrow \quad d^{(m)} = m\cdot\left[1-(1-d)^{1/m}\right] = m\cdot\left[1-(1+i)^{-1/m}\right],$$

   then by substitution for $d^{(m)}$,

$$\ddot{s}_{\overline{n}|}^{(m)i} = \frac{(1+i)^n - 1}{d^{(m)}}.$$

4. As
$$\ddot{a}^{(m)}_{\overline{n}|} = v^n \, \ddot{s}^{(m)}_{\overline{n}|},$$
it follows immediately from the first part that
$$\ddot{a}^{(m)}_{\overline{n}|} = \frac{1 - v^n}{i^{(m)}}.$$

Solution 3 Either using the formula developed in the earlier questions or by changing the time unit:

1. $s^{(2)\,0.05}_{\overline{10}|} = \frac{1}{2}\, s^{0.024695}_{\overline{20}|} = 12.73320.$
2. $a^{(6)\,0.1}_{\overline{17}|} = \frac{1}{6}\, a^{0.016012}_{\overline{102}|} = 8.34959.$
3. $\ddot{s}^{(4)\,0.035}_{\overline{25}|} = \frac{1}{4}\, \ddot{s}^{0.008637}_{\overline{100}|} = 39.79822.$
4. $\ddot{a}^{(12)\,0.12}_{\overline{7}|} = \frac{1}{12}\, \ddot{a}^{0.009489}_{\overline{84}|} = 4.85527.$
5. As $v^n \to 0$ as $n \to \infty$,
$$\ddot{a}^{(m)}_{\overline{\infty}|} = \lim_{n\to\infty} \ddot{a}^{(m)}_{\overline{n}|} = \lim_{n\to\infty} \frac{1 - v^n}{d^{(m)}} = \frac{1}{d^{(m)}} - \lim_{n\to\infty} \frac{v^n}{d^{(m)}} = \frac{1}{d^{(m)}},$$
we get $\ddot{a}^{(4)\,0.06}_{\overline{\infty}|} = 1/d^{(4)} = 17.28711$. Alternatively,
$$\ddot{a}^{(4)\,0.06}_{\overline{\infty}|} = \frac{1}{4}\, \ddot{a}^{0.014674}_{\overline{\infty}|} = 17.28711.$$